

OBJECT ORIENTED PROGRAMMING USING JAVA

Time Allowed: 2.5 Hours

Full Marks: 60

Answer to Question No. 1 of Group A must be written in the main answer script.
In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.
Answer any Five (05) Questions from Group-B.

GROUP-A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): 2x10=20

i) What does the JAVA Runtime Environment (JRE) include?

- a) Java Compiler only
- b) Tools for Java application deployment
- c) Integrated Development Environment (IDE)
- d) Java Virtual Machine (JVM), Java class libraries and the Java class loader.

ii) Which of the following is not a java feature?

- a) Use of explicit pointers
- b) Robust
- c) Distributed
- d) Portable

iii) Which of the following type of inheritance, java does not support?

- a) Multilevel
- b) Hierarchical
- c) Single
- d) Multiple

iv) Which of the following keyword is used for implementing an inheritances in Java?

- a) implements
- b) extends
- c) super
- d) None of these

v) Which is of the following is a valid declaration of an object obj of class Room?

- a) Room() obj = new Room();
- b) Room obj = new Room();
- c) Room obj = new Room;
- d) obj = new Room();

vi) Which of the following keyword is used for implementing an interface in Java?

- a) implements
- b) extends
- c) final
- d) None of these

vii) Which keyword is used for accessing the features of a package?

- a) export
- b) package
- c) import
- d) extends

viii) Which of the following modifiers can not be used for constructor?

- a) private
- b) public
- c) static
- d) protected

ix) Which of the following is not an access modifiers ?

- a) void
- b) public
- c) private
- d) protected

x) What is the extension of compiled java classes?

- a) .java
- b) .class
- c) .txt
- d) .obj

xi) What is the number of bytes used by java primitive data type boolean?

- a) 1
- b) 4
- c) 8
- d) 2

xii) Given a one dimensional array arr, what is the correct way of getting its number of elements?

- a) arr.size
- b) arr.length-1
- c) arr.length
- d) arr.size-1

xiii) Which of this not a built in package?

- a) java.awt
- b) java.io
- c) java.lang
- d) All of these

xiv) Which of the following is a mutable class in Java?

- a) java.lang.String
- b) java.lang.Byte
- c) java.lang.Short
- d) java.lang.StringBuilder

- xv) Which of these class is used to read and write bytes in a file?
a) FileReader b) FileWriter c) ByteStream d) InputstreamReader

GROUP-B
Answer any Five (05) questions.

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| 2. | a) Explain any three features of java programming.
b) What do you mean byte code?
c) Why java is not considered as a completely object-oriented programming language? | 4 + 2 + 2 |
| 3. | a) What is type casting? Explain its types with proper syntax and example.
b) What do you understand by Garbage Collection?
c) Write a java code to find the sum of digits of a number. | 3 + 2 + 3 |
| 4. | a) Define class and object with example
b) Explain the meaning of each word of the following line:
<code>public static void main(String args [])</code> | (2 + 2) + 4 |
| 5. | a) Explain Constructor.
b) Write a Java Code to implement constructor overloading
c) State the purpose of using final keyword with example | 3 + 3 + 2 |
| 6. | a) What is inheritance?
b) Write a Java Code to implement multilevel inheritance.
c) Differentiate between method overloading and method overriding. | 1 + 4 + 3 |
| 7. | a) What is meant by interface?
b) Write a java code to implement multiple inheritance using interface.
c) Write the difference between interface and class. | 1 + 4 + 3 |
| 8. | a) Define package with example.
b) Explain try, catch and throw block used in java Exception Handling. | 2 + 6 |
| 9. | a) State different types of errors.
b) Describe thread synchronization in respect to multithreading.
c) Define thread priority. | 2 + 4 + 2 |
| 10. | a) Define stream class. b) State the classification of stream classes.
c) Write a java program to create and read text file. | 1 + 2 + 5 |
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